

Program: Diploma in Artificial Intelligence & Machine Learning	
Course Code : 5342	Course Title: Soft Computing
Semester: 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To become familiar with neural networks and understand their features and applications.
- To get exposed to the mathematical background for solving problems using neural network learning algorithms.
- To get exposed to the ideas of fuzzy logic and also provide knowledge about the use of inference systems and approximate reasoning in solving complex tasks.
- To familiarise with genetic algorithms useful for solving complex optimisation.

Course Prerequisites:

Topic	Course code	Course name	Semester
AI Problems	3344	Artificial Intelligence	3
Artificial Neural Networks	4342	Machine Learning & Neural Networks	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize fundamental concepts of soft computing	10	Understanding
CO2	Explain the concepts of Artificial Neural Network (ANN)	16	Understanding
CO3	Demonstrate fuzzy logic and fuzzy sets	16	Understanding
CO4	Understand the concepts of genetic algorithms	16	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3	3		3			
CO3	3	3		3			
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Descriptions	Duration (Hours)	Cognitive Level
CO1	Summarise fundamental concepts of soft computing		
M1.01	Familiarise soft computing Concepts	2	Understanding
M1.02	Differentiate "Soft" computing and "Hard" computing	2	Understanding
M1.03	Discuss conventional AI to Computational Intelligence.	2	Understanding
M1.04	Familiarize feed forward, adaptive networks.	2	Understanding
M1.04	Applications of Soft computing techniques	2	Understanding
Contents: Basic concepts of soft computing -Definition, Elements of soft computing, Characteristics of Soft Computing. Differentiate Soft Computing and Hard Computing, Conventional AI, Computational Intelligence, Feed-forward networks, Adaptive networks, Applications of Soft computing.			
CO2	Outline concepts of Artificial Neural Network (ANN)		
M2.01	Understand Artificial Neural Networks.	2	Understanding
M2.02	Outline the basic models of ANN.	5	Understanding
M2.03	Summarise important terminologies of ANN	5	Understanding
M2.04	Understand basic networks in Supervised Learning	4	Understanding

Contents: Artificial Neural Networks: Biological foundations –ANN models - Characteristics of ANN - Types of activation function Important terminologies of ANN – weights, bias, threshold, learning rate, momentum factor, vigilance parameter, notations-McCulloch-Pitts Neuron - overfitting and underfitting Single layer, Multilayer, Recurrent networks.Supervised and unsupervised learning Overview of Supervised Learning Network- Perceptron Network- Adaline - Madaline			
CO3	Demonstrate fuzzy logic and sets		
M3.01	Familiarise fuzzy sets	2	Understanding
M3.02	Illustrate fuzzy Operations and Properties	5	Understanding
M3.03	Summarize Fuzzy Relations and Membership functions	5	Understanding
M3.04	Understand Fuzzy Rules and Fuzzy logic	4	Understanding
Contents: Introduction to Fuzzy Sets - Properties Basic fuzzy set operations Fuzzy Relations - Cardinality - Operations on fuzzy relations. Membership functions- Features of the membership functions., Fuzzy Rules and Fuzzy Reasoning Fuzzy Inference Systems, Fuzzy Logic, Fuzzy Expert Systems			
CO4	Understand Concepts of genetic algorithms		
M4.01	Define Genetic algorithm	1	Understanding
M4.02	Understand Biological terminologies	3	Understanding
M4.03	Understand basic terminologies and workings of genetic algorithms.	3	Understanding
M4.04	Explain Simple and General Genetic algorithm	3	Understanding
M4.05	Illustrate Operators in Genetic Algorithm	4	Understanding
M4.06	Outline Applications of Genetic Algorithm	2	Understanding
Contents: Genetic Algorithms(GA)-Basic biological terminologies, Terminologies and working of genetic algorithms. Simple genetic algorithm. General Genetic Algorithm, Operators in GA –encoding- selection- recombination- mutation. Applications of Genetic Algorithms.			

Text / Reference

T/R	Book Title/Author
T1	S.N. Sivanandam, S.N. Deepa, “ Principles of Soft Computing”, Wiley India, 2nd Ed.,2011.
T2	Jyh-Shing Roger Jang, Chuen-Tsai Sun, Eiji Mizutani, ”Neuro-Fuzzy and Soft Computing: A Computational Approach to Learning and Machine Intelligence, Prentice Hall, Upper Saddle River, NJ, ©1997.
T3	Mitchell Melanie, “An Introduction to Genetic Algorithms”, MIT Press, 1998.
T4	B. Yegnanarayana, “Artificial Neural Networks”, PHI, 1999.
R1	Laurene Fauseett: ”Fundamentals of Neural Networks”, Prentice Hall India, New Delhi,1994.
R2	David A. Coley, “Introduction to Genetic Algorithms for Scientists and Engineers”, World Scientific Publishers, 1999

Online Resources

Sl.No	Website Link
1	https://www.elprocus.com/soft-computing/
2	https://www.tutorialspoint.com/adaline-and-madaline-network
3	https://www.turing.com/kb/genetic-algorithm-applications-in-ml